Jakkala Naga Rohith Kumar

cs23btech11021@iith.ac.in | nagarohithkumarjakkala@gmail.com | +91 9392272525 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
Bhasyam Junior College, Gorantla, Guntur	Class XII (Board of Intermediate Education AP)	2023	96.70%
Bhasyam HS NR Peta Kurnool	Class X (Board of Secondary Education AP)	2021	10.00%

WORK EXPERIENCE

Oracle Hyderabad -- Application Developer Intern

- Contributed to **HILO**, an agentic AI Talent Advisor application built on Oracle AI Agent Studio, enabling managers and HRs to access talent-related insights and guided decision-making through conversational workflows.
- Designed and configured agentic workflows for HILO using platform-native patterns, improving interaction handling for manager and HR use cases.
- Built a **BugDB analytics dashboard** for managers, featuring interactive visualizations, which helps managers to monitor current progress of the department they are working and individual employee under them and helps in bugs tracking.

PROJECTS

Observability Platform for Logs and Exceptions

- Designed and implemented a scalable observability platform for real-time ingestion, processing, and analysis of logs, metrics, and traces using Envoy, Vector, Redpanda, Flink, and ClickHouse**.**
- Built hot/cold telemetry pipelines with intelligent routing, anomaly detection, tail sampling, error-rate aggregation, and Grafana/Prometheus dashboards for real-time monitoring.

AFT - Domain-Specific Language for Signal Processing: GitHub Repo

- Developed a statically typed domain-specific language for signal processing, supporting complex numbers, vectors, functions, control flow, and structs, with built-in FFT/IFFT, digital filtering, correlation, resampling, and spectral analysis capabilities.

RISC-V Simulator using Imgui: Github Repo

- Designed and implemented a RISC-V simulator GUI in C++, supporting instruction execution, memory access, and cache behavior visualization for low-level systems analysis for the course Computer Architecture.
- used Imgui for GUI, native file dialog for file handling

Inventory Management System: Github Repo

- Designed and implemented a responsive web-based Inventory Management System to handle inventory tracking, billing, and historical analytics as the part of course Database managament systems.
- used nextjs for frontend, go for backend and mysql for database

Sudoku Validation using threads : Github Repo

- Designed and tested multiple thread-based Sudoku validation methods in C++, conducting performance analysis to compare execution efficiency and speed across varying strategies, including thread count and input size variations as the part of OS course.

GemOs Memory Allocation: Github Repo

- Implemented memory allocation routines in gemOS as part of an academic assignment in an Operating Systems course, to learn and apply core concepts of virtual memory, paging, and low-level memory management in C as part of OS course.

Chess game in rust: Github Repo

- Designed and Implement chess game in rust using Macroquad library which focuses on game logic and library functions implementation.

SKILLS

Languages: C++, C, Python, Rust, JS, HTML, Go

Python Libraries: Numpy, Pandas, Scikit-learn, Pytorch, Matplotlib, Networkx

Database Skills: MySQL, PostgreSQL, MongoDB

Systems/Low-level: OOP, RISC-V, Cache Architecture, Virtual Memory

Dev Tools: Git, Docker, VS Code, vim, Bash, LaTeX, npm

AI : Langchain, Litellm, Ollama, tools, MCP, Prompt Engineering, Context Engineering

Web Development: React, NextJs, JavaScript, HTML, CSS, Tailwindcss, Flask

Environments: Linux, MacOS, Windows

POSITIONS OF RESPONSIBILITY

Teaching Assistant: Course - **Computer Architecture**

- Worked as a **Teaching Assistant** under **Prof. Rajesh Kedia** for the **Computer Architecture** course, which focuses on processor design, ISA, RISCV, memory hierarchy, pipelining, cache organization, and performance optimization.

Teaching Assistant : Course- **Intro to Programming** .

- worked as **Teaching Assistant** under **Prof. Jyothi Vedurada** for the **Intro to programming** course, which mainly focuses on C language and basic Algorithms

RELEVANT COURSES

Computer Science: Database Management System, Operating System, Computer Architecture, Computer Networks, Compilers, Software Engineering, Data Structures, Algorithms, Quantum Computing, Theory of computation

AI : Distributed AI Training, Multi-Agent Systems, Linear Optimization, Deep Learning, Foundations of Machine Learning

EXTRACURRICULAR

- Part of NSS
- Reached pupil in codeforces with a rating of 1304 (codeforces profile)
- reached 3 star in codechef platform with rating of 1686 (codechef profile)